Lab 06
Differential Amplifier

Analog IC Design Course – Cadence Tools (Self Study)

Student Information

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Under Guidance and Help of My More Experienced Friends

Labs Provided by **Dr. Hesham Omran** via Master-Micro.com

Thanks to everyone who has helped me,
I pledge to help others as well.

Done during my **Post Graduate MSc. Program** in
Electronics and Electrical Communication Engineering
at the **Faculty of Engineering – Ain Shams University**

Lab 06: Differential Amplifier

Intended Learning Objectives:[1]

In this lab[2] you will:

- Design and simulate a differential amplifier.
- Learn how to simulate the small-signal differential characteristics of a differential amplifier.
- Learn how to simulate the small-signal common-mode characteristics of a differential amplifier.
- Learn how to simulate the large-signal differential characteristics of a differential amplifier.
- Learn how to simulate the large-signal common-mode characteristics of a differential amplifier.

Part 1: Sizing Chart

1) We can show that the intrinsic gain of a MOSFET is given by: $|Av| \approx g_m r_o = \frac{2I_D}{V_{ov}} \times \frac{V_A}{I_D} = \frac{2V_A}{V_{ov}}$

Interestingly, the gain only depends on λ and V_{ov}. However, to derive this expression we used $g_m = \frac{2I_D}{V_{ov}}$ which is based on the square-law. For a real MOSFET, if we compute V_{ov} and $\frac{2I_D}{g_m}$ they will not be equal. Let's define a new parameter called V-star (V^*) which is calculated from actual simulation data using the formula $V^* = \frac{2I_D}{g_m} \leftrightarrow g_m = \frac{2I_D}{V^*}$

For a square-law device, $V^* = V_{ov}$, However, for a real MOSFET they are not equal. The actual gain is now given by: $|Av| \approx \frac{2V_A}{V^*}$

The lower the V^* the higher the gain, but the larger the area and the lower the speed.

2) We want to design a differential amplifier with the specifications below. Note that the bias current is split between two transistors; each transistor gets $\boldsymbol{I_D = 20\mu A}$.

Parameter	130 nm CMOS	180 nm CMOS
L	$\mathbf{1\ \mu m}$	$\mathbf{1\ \mu m}$
CM output level [3]	**0.5**	**0.7**
Differential gain	**5**	**8**
Supply	$\mathbf{1.2\ V}$	$\mathbf{1.8\ V}$
Bias current I_{ss}	$\mathbf{40\ \mu A}$	$\mathbf{40\ \mu A}$

$$Put\ L = 1\ \mu m$$

Again, I decided – in this lab – to use the **umc13mmrf** PDK (Process Design Kit), which is a 130nm process technology, instead of the **ee214b.sp** file. The reason is a matter of preference.

[1] *COLOR CODES:* Orange (steps) -- Blue (output) -- Green (comments / observations) – White (equations).

[2] **Cadence Virtuoso** software on a Linux® Debian 8 Virtual Machine is used.

[3] **ISS*RD** must be smaller than ($\boldsymbol{V_{DD} - V_{dsat3}}$) for proper large signal characteristics (why?).

ISS*RD must be smaller than ($\boldsymbol{V_{DD} - V_{dsat3}}$) for proper large signal characteristics (why?).

$$@\max swing:\ V_{out,1} = I_{SS}R_D, \qquad V_{out,2} = 0$$
$$V_{out,MAX\ SWING} = V_{out,1} - V_{out,2} = I_{SS}R_D$$

$$for\ M1\ to\ be\ in\ saturation:\ V_{DG1} \le |V_t|$$
$$V_{out1} - V_{iCM} \le |V_t|$$
$$V_{out,Max} = V_{iCM} + |V_t|$$

$$V_{out,DIFF\ Swing} = V_{odiff_{max}} - V_{odiff_{min}} = I_{SS}R_D - (-I_{SS}R_D) = 2I_{SS}R_D = 2(V_{out,1} - V_{out,2})$$
$$= 2(V_{iCM} + |V_t|) = 2(V_{DD} - \boldsymbol{V_{SG1}} - V_3^* + |\boldsymbol{V_t}|) = 2(V_{DD} - \boldsymbol{V_1^*} - V_3^*)$$

$$2I_{SS}R_D = 2(V_{DD} - V_1^* - V_3^*)$$
$$I_{SS}R_D = V_{DD} - V_3^* - V_1^*$$

$$\therefore I_{SS}R_D < V_{DD} - V_3^*$$

3) Choose $\boldsymbol{R_D}$ to meet the CM output level spec.

4) We can show that the differential amplifier gain is given by $|\boldsymbol{A_v}| \approx \boldsymbol{g_m R_D} = \frac{\boldsymbol{2I_D}}{\boldsymbol{V^*}} \times \boldsymbol{R_D} = \frac{\boldsymbol{2V_{R_D}}}{\boldsymbol{V^*}}$

5) Choose $\boldsymbol{V^*}$ to meet the differential gain spec.

$$|\boldsymbol{A_v}| \approx \boldsymbol{g_m R_D} = \frac{\boldsymbol{2I_D}}{\boldsymbol{V^*}} \times \boldsymbol{R_D} = \frac{\boldsymbol{2V_{R_D}}}{\boldsymbol{V^*}} = \boldsymbol{5}$$
$$\boldsymbol{V_{out,CM}} = \boldsymbol{V_{R_D}} = I_D R_D = 0.5$$

$$\boldsymbol{R_D} = \frac{0.5}{I_D} = \frac{0.5}{20\ \mu A} = \boldsymbol{25\ k\Omega}$$

$$\frac{\boldsymbol{2V_{R_D}}}{\boldsymbol{V^*}} = \boldsymbol{5}$$

$$\frac{2(0.5\ V)}{V^*} = 5$$

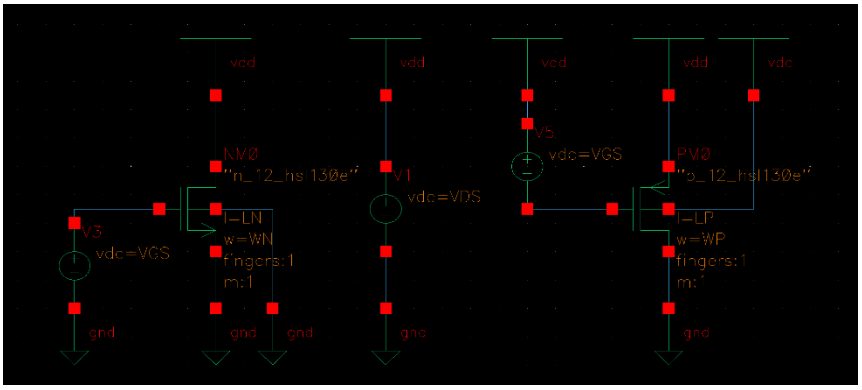
$$\boldsymbol{V^* = 200mV}$$

6) The remaining variable in the design is to calculate $\boldsymbol{W}$. Since the square-law is not accurate, we cannot use it to determine the sizing. Instead, we will use a sizing chart generated from simulation.

Create a test-bench for PMOS transistor as shown below (we will use PMOS only in this lab). Use $\boldsymbol{W}$ = 10 $\boldsymbol{\mu m}$ (we will understand why shortly).

7) Sweep VGS from $0\ to\ \approx V_{TH} + 0.4V$ with 10mV step. Set $V_{DS} = V_{DD}/2$.

$$V_{TH} + 0.4V \approx 0.87\ V\ [let\ it\ be\ 1V], \qquad V_{DD} = 1.2\ V, \qquad V_{DS} = 0.6\ V$$



8) We want to compare $V^* = 2I_D/g_m$ and $V_{ov} = V_{GS} - V_{TH}$ by plotting them overlaid. Use the calculator to create expressions for V^* and V_{ov}. You can save the expressions to reuse them later.

The Blue Curve is $V_{ov} = V_{GS} - V_{TH}$

```
(v("NM0:vgs" ?result "dc") - v("NM0:vth" ?result "dc"))
```

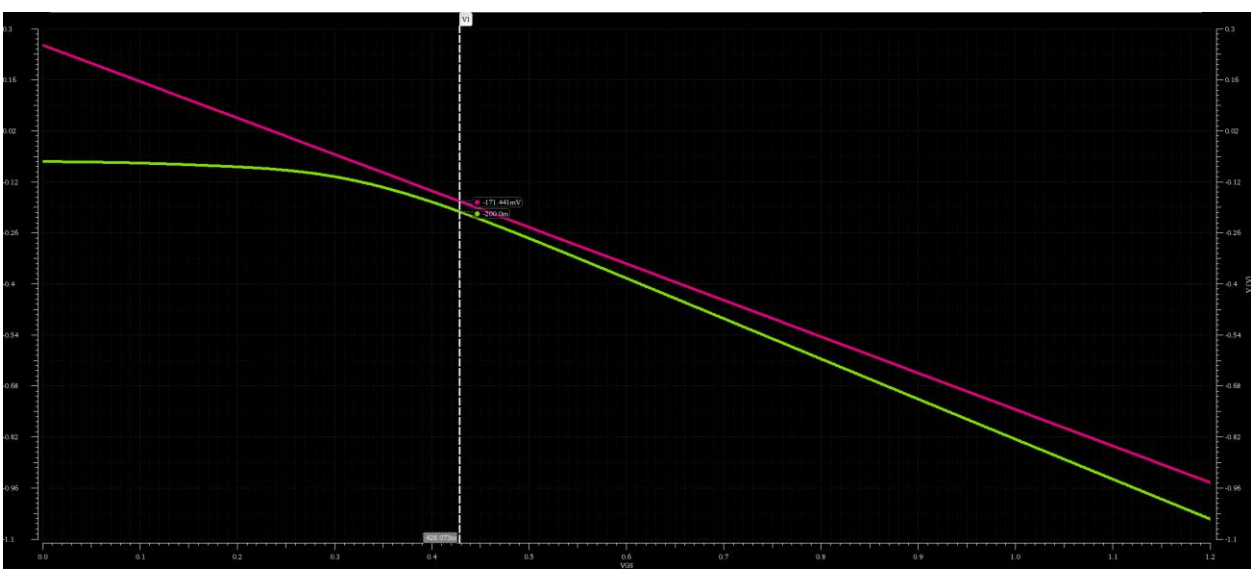
The Green Curve is $V^* = 2I_D/g_m$

```
(2 / (getData("NM0:gm" ?result "dc") / getData("NM0:id" ?result "dc")))
```

The difference is that V^* takes a lot of factors into consideration, unlike the simple V_{ov} from square law.

9) Plot V^* and V_{ov} overlaid vs. V_{GS}. Make sure the y-axis of both curves has the same range. You will notice that at the beginning of the strong inversion region, V^* and V_{ov} are relatively close to each other (i.e., square-law is relatively valid). For deep strong inversion (large V_{ov}: velocity saturation and mobility degradation) or weak inversion (near-threshold and sub-threshold operation) the behavior is quite far from the square-law.

The y-axes ranges are (-1.1V to 0.3V) and x-axis is (0 to 1.2V). For PMOS



10) On the V^* and V_{ov} chart locate the point at which V^* is equal to the value your previously calculated to meet the gain spec. Find the corresponding V_{ovQ} and V_{GSQ}.

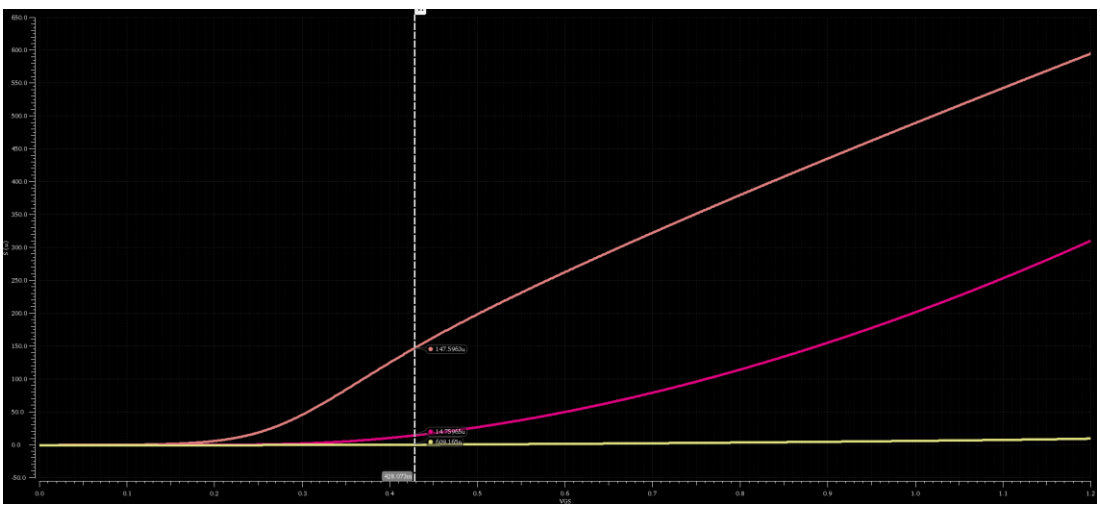
We chose $V^* = 200\ mV$ as per given in specs.

PMOS
$V^* = V_Q^* = -200\ mV$
$V_{ov} = V_{ov_Q} = -171.441\ mV$
It is noted that $V_{GS} = V_{GS_Q} = 428.073\ mV$

11) Plot I_D, g_m, and g_{ds} vs. V_{GS}. Find their values at V_{GS_Q}. Let's name these values I_{DX}, g_{mX}, and g_{dsX}.

$I_{DX} = -14.76\ \mu A$
$g_{mX} = 147.6\ \mu S$
$g_{dsX} = 609.2\ nS$
$r_o = 1.642\ M\Omega$

getData("/PM0/S" ?result 'dc)	
vgs_p	-428.1m
vth_p	-256.6m
id_p	-14.76u
gm_p	147.6u
gds_p	609.2n
Vov_p	
(2ID/Gm)_p	
ro_p	1.642M



12) Now back to the assumption that we made that $W = 10\mu m$. This is not the actual value that we will use for our design. But the good news is that I_D is always proportional to W irrespective of the operating region and the model of the MOSFET (regardless square-law is valid or no). Thus, we can use ratio and proportion (cross-multiplication) to determine the correct width at which the current will be $I_{DQ} = 10\mu A$ as given in the specs. Calculate W as shown below.

Wn	I_D
10 µm	I_{DX} @ V_Q^* (from the chart)
?	$I_{DQ} = 20\ \mu A$ (from the specs)

$$W = \frac{10\ \mu m \times 20\ \mu A}{I_{DX}\ @\ V_Q^*\ \text{(from the chart)}}$$

$$W_{pmos} = \frac{10\ \mu m \times 20\ \mu A}{\mathbf{14.75965\ \mu A}} = 13.55045\ \mu m \approx \mathbf{14\ \mu m}$$

13) Now we are almost done with the design of the amplifier. Note that $\boldsymbol{g_m}$ is also proportional to $\boldsymbol{W}$ as long as $\boldsymbol{V_{ov}}$ is constant. On the other hand, $\boldsymbol{r_o = 1/g_{ds}}$ is inversely proportional to $\boldsymbol{W}$ $(\boldsymbol{I_D})$ as long as L is constant. Before leaving this part, calculate $\boldsymbol{g_{mQ}}$ and $\boldsymbol{g_{dsQ}}$ using ratio and proportion (cross multiplication).

g_{mQ}	W @ constant V_{ov}
$\mathbf{147.6\ \mu S}$	$10\ \mu m$
?	$8\ \mu m$

g_{dsQ}	W (I_D)
$\mathbf{609.2\ nS}$	$10\ \mu m$
?	$14\ \mu m$

$$\boldsymbol{g_{mQ}} = \frac{\mathbf{147.6\ \mu S} \times 14\ \mu m}{10\ \mu m} = 206.64\ \mu S$$

$$\boldsymbol{g_{dsQ}} = \frac{\mathbf{609.2\ nS} \times 14\ \mu m}{10\ \mu m} = 852.88\ nS$$

$$r_o = \frac{1}{852.88\ nS} \approx 1.1725\ M\Omega$$

<u>Note:</u> $g_m = \mathbf{147.6\ \mu S} \rightarrow g_m = \mathbf{206.64\ \mu S}$
Directly proportional to W (increased)

$r_o = \mathbf{1.6415\ M\Omega} \rightarrow r_o \approx \mathbf{1.1725\ M\Omega}$
Inversely proportional to W (decreased)

REVIEW ON OUR **FINAL** PARAMETERS:

$$\therefore L = 1\ \mu m, \qquad W = 14\ \mu m, \qquad V_{DD} = 1.2\ V,$$

$$\boldsymbol{V^* = V_Q^* = -200\ mV}$$
$$\boldsymbol{V_{ov} = V_{ov_Q} = -171.441\ mV}$$
$$\boldsymbol{V_{GS} = V_{GS_Q} = 428.073\ mV}$$

$$\therefore g_m = 206.64\ \mu S, \qquad g_{ds} = 852.88\ nS, \qquad r_o = 1.1725\ M\Omega$$

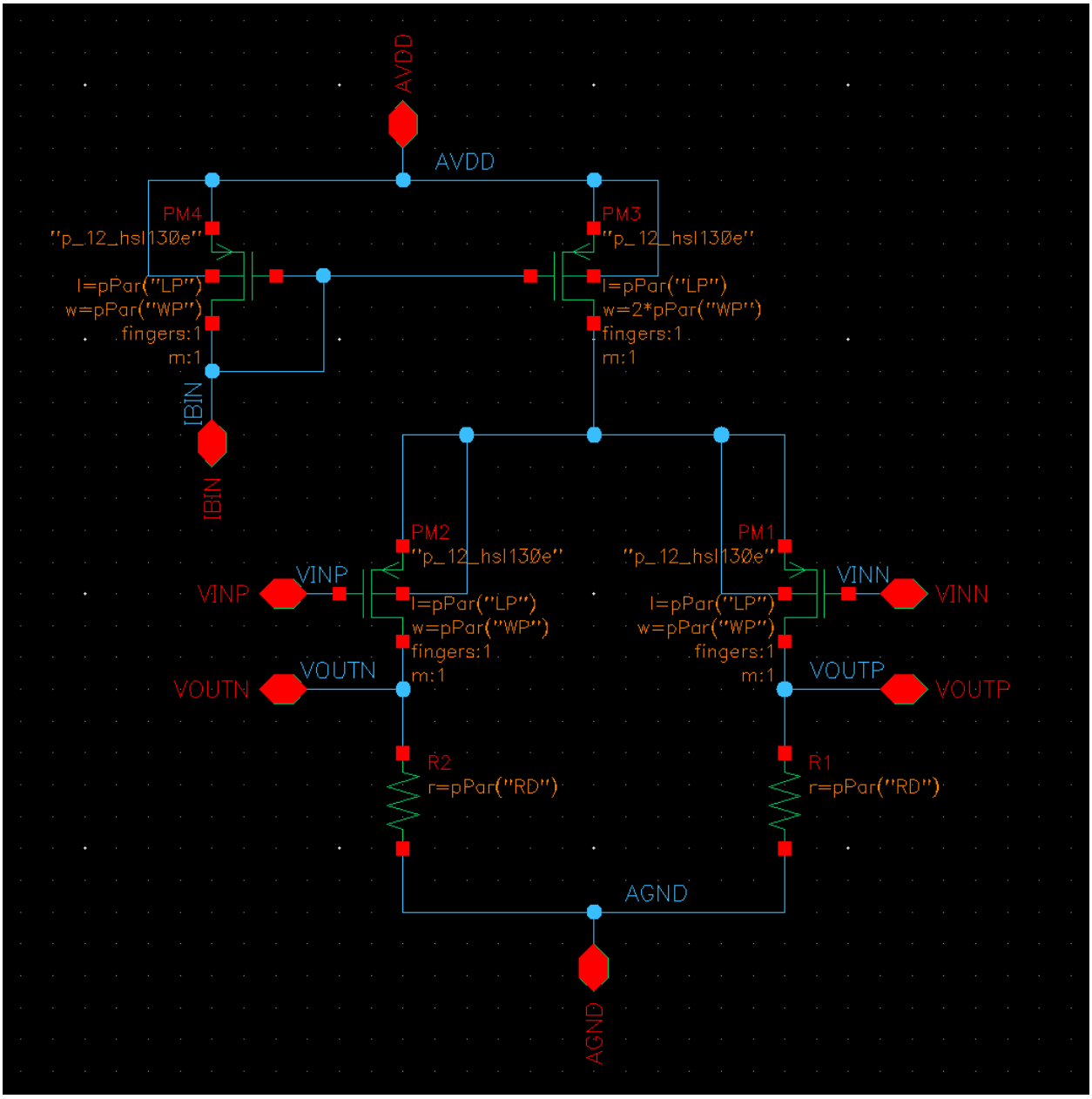
Simulator Results:

vgs_p	-428.1m
vth_p	-256.7m
id_p	-20.69u
gm_p	207u
gds_p	853.8n
Vov_p	
(2ID/Gm)_p	
ro_p	1.171M

END OF SIZING AND DESIGN

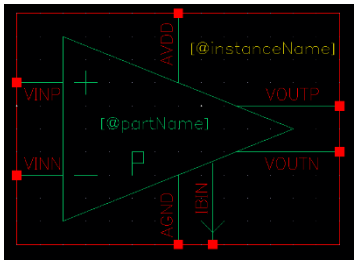
Part 2: Differential Amplifier

1) Create a new cell for the diff amp "**lab_06_diff_amp**". Create the schematic of a *differential amplifier with PMOS input stage and resistive load*. Use a simple current mirror for the bias current source as shown in the first figure in this document.



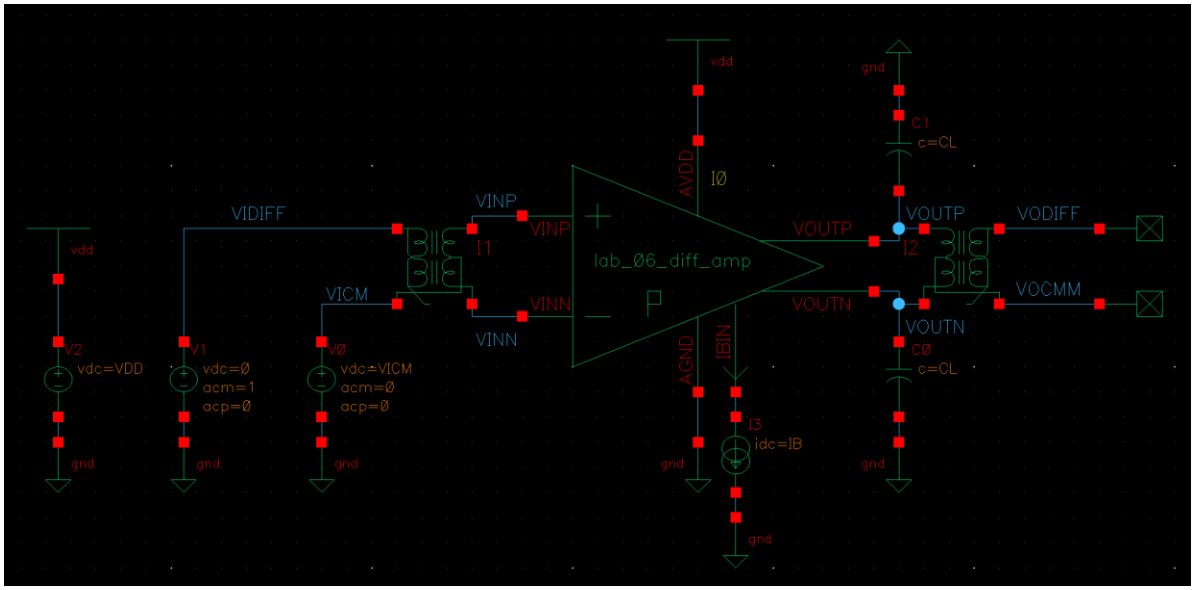
2) Create a symbol for the diff pair
(use Create -> Cellview -> From Cellview).

3) Edit the diff amp symbol to be as shown below.



4) Create a new cell for the testbench "**lab_06_diff_amp_tb**". Create testbench schematic as shown below.

➔ **Cadence Hint:** Use ***ideal_balun*** from ***analogLib*** to separate common-mode and differential signals. In Mentor tools you have to create the balun yourself.



5) Size the transistors as designed in **Part 1**.

$$\therefore \boldsymbol{L = 1\ \mu m}, \qquad \boldsymbol{W = 14\ \mu m}$$

6) Use **IB = 20uA** (the CM multiplies this by 2, so each half in the diff pair gets **IB**).
7) Connect floating wires to ***noConn*** instance from basic library to avoid floating signal warnings.
8) Use **CL = 1pF**.

9) Analytically calculate the valid range for **Vicm**: the common mode input range (**CMIR**). Set **Vicm** at the center of this range.

➔ **Cadence Hint:** In adexl you can access the results of the last 10 simulation runs from the "History" tab. We want to access the last two simulation runs in order to compare **Avd** and **Avcm**. Open both simulations in the Results Browser.

$\boldsymbol{M3\ in\ sat.}$:

$V_{SD3} \geq V_3^*$

$V_{DD} - V_X \geq V_3^*$

$V_X = V_{iCM} + V_{SG2}$

$V_{DD} - (V_{iCM} + V_{SG2}) \geq V_3^*$

$\boldsymbol{V_{DD} - V_{SG2} - V_3^* \geq V_{iCM}}$

$\boldsymbol{M1\ in\ sat.}$:

$V_{DG1} \leq |V_t|$

$I_D R_D - V_{iCM} \leq |V_t|$

$I_D R_D - |V_t| \leq V_{iCM}$

$\boldsymbol{V_{iCM} \geq I_D R_D - |V_t|}$

$$\boldsymbol{I_D R_D - |V_t| \leq V_{iCM} \leq V_{DD} - V_{SG2} - V_3^*}$$

$$0.5V - 256.7\ mV \leq \boldsymbol{V_{iCM}} \leq 1.2V - 428.073\ mV - 0.2V$$

$$\boldsymbol{CMIR} \rightarrow 243.3\ mV \leq \boldsymbol{V_{iCM}} \leq 571.927\ mV$$

$$Put\ \boldsymbol{V_{iCM}}\ in\ the\ center\ of\ this\ range = \boldsymbol{407.6135\ mV}$$

Report the following:

1) OP simulation.

- Report the schematic of the diff pair with DCOP point clearly annotated: id, vgs, vds, vth, vdsat, gm, gds, gmb, region.
- Check that all transistors operate in saturation.

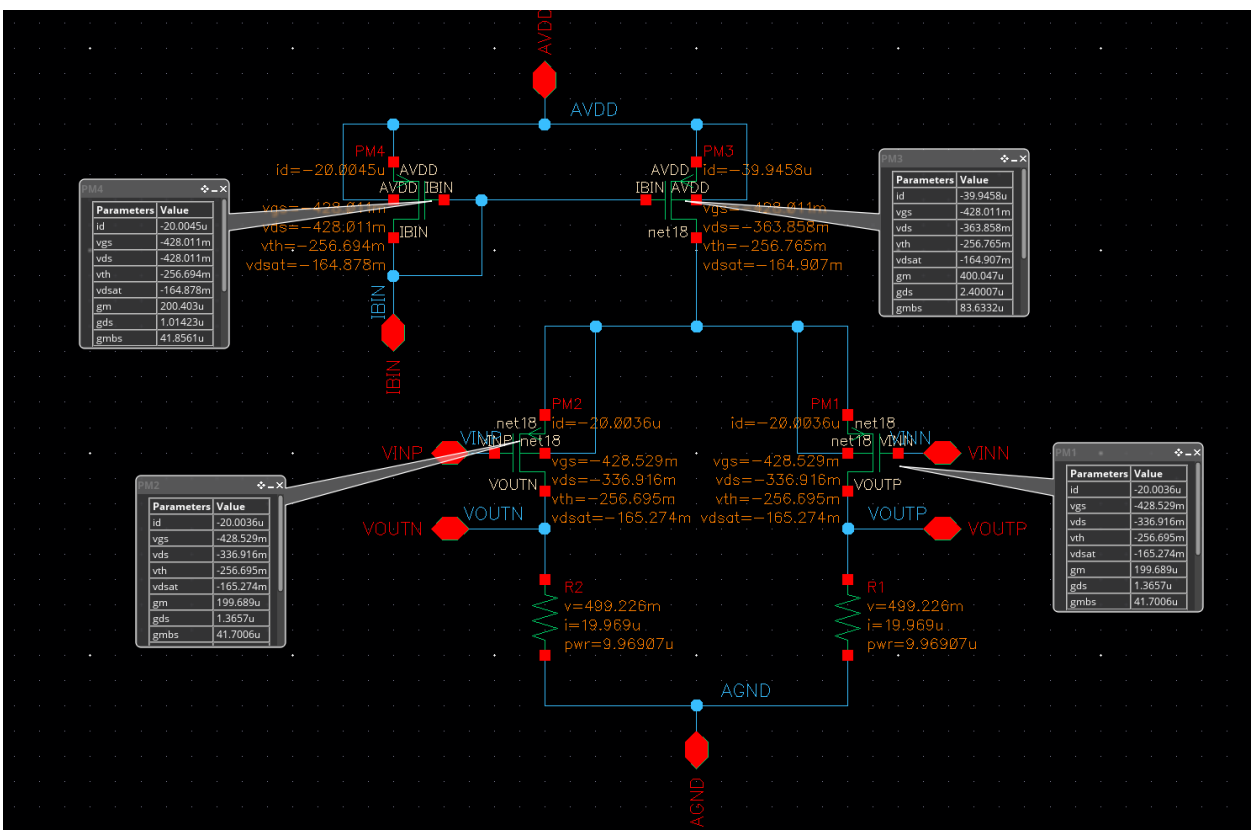
id_PM1	-20u
vgs_PM1	-428.5m
vds_PM1	-336.9m
vth_PM1	-256.7m
vdsat_PM1	-165.3m
gm_PM1	199.7u
gds_PM1	1.366u
gmb_PM1	41.7u
region_PM1	2

id_PM2	-20u
vgs_PM2	-428.5m
vds_PM2	-336.9m
vth_PM2	-256.7m
vdsat_PM2	-165.3m
gm_PM2	199.7u
gds_PM2	1.366u
gmb_PM2	41.7u
region_PM2	2

id_PM3	-39.95u
vgs_PM3	-428m
vds_PM3	-363.9m
vth_PM3	-256.8m
vdsat_PM3	-164.9m
gm_PM3	400u
gds_PM3	2.4u
gmb_PM3	83.63u
region_PM3	2

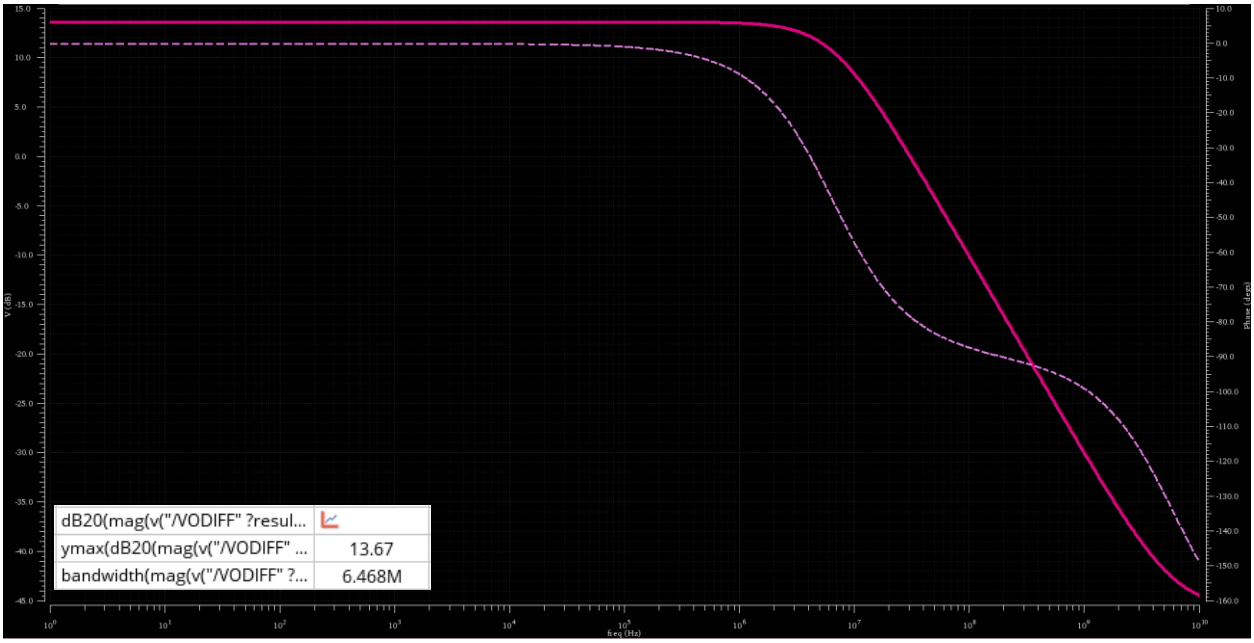
id_PM4	-20u
vgs_PM4	-428m
vds_PM4	-428m
vth_PM4	-256.7m
vdsat_PM4	-164.9m
gm_PM4	200.4u
gds_PM4	1.014u
gmb_PM4	41.86u
region_PM4	2

All Transistors Operate in Saturation.



2) Diff small signal ccs. :

- Use **AC magnitude = 1** for the **diff** source (and **AC magnitude = 0** for the **CM** source).
- Set **Vicm** at the center of the CMIR.
- Run **AC analysis** (1Hz: 10GHz, logarithmic, 10 points/decade).
- Report the Bode plot of **small signal diff gain**.
- Compare the **DC diff gain and BW** with **hand analysis** in a table.



Hand Analysis (A1):

$$A_{vd} = \frac{v_{od}}{v_{id}} = \frac{-g_m R_D (v_{id})}{v_{id}} = -g_m R_D = A_{v,SE}$$

$$A_{vd} = -g_m R_D = -(199.689\mu)(25k) = -4.9922$$

$$20\log|A_{vd}| = 13.97dB$$

$$\omega \approx \frac{1}{R_{out}C_{out}} = \frac{1}{(r_o||R_D)(C_L)} = \frac{1}{25k \times 1p} = 40 \times 10^6$$

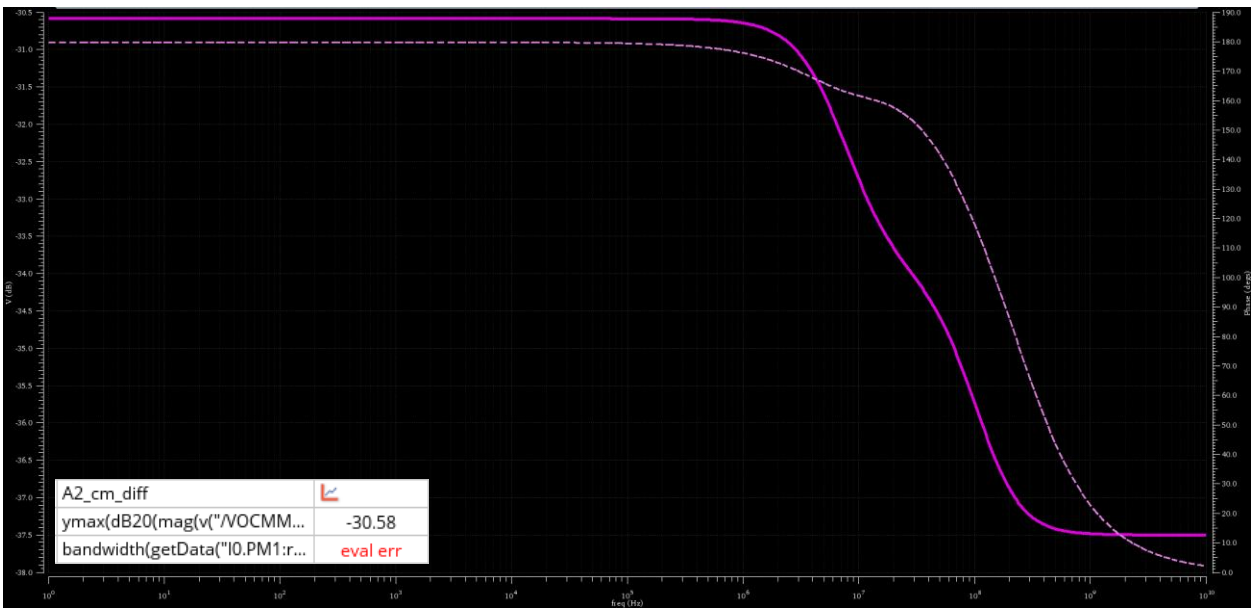
$$f = \frac{40 \times 10^6}{2\pi} = 6.3662\ MHz$$

	Simulation	Hand Analysis	Relative Error (ϵ)
DC diff gain	13.67 dB	13.97 dB	-2.14 %
BW	6.468 MHz	6.3662 MHz	1.6 %

3) CM small signal ccs. :

- Use **AC magnitude = 1** for the **CM** source (and **AC magnitude = 0** for the **diff** source).
- Run **AC analysis** (1Hz: 10GHz, logarithmic, 10 points/decade).
- Report the Bode plot of **small signal CM gain**.
- Compare the **DC CM gain** with **hand analysis** in a table. Is it smaller than "1"? Why?
- Justify the variation of **Avcm vs. frequency**.
- Plot **Avd/Avcm** in dB. Compare **Avd/Avcm @ DC** with **hand analysis** in a table.
- Justify the variation of **Avd/Avcm with frequency**.

➔ **Cadence Hint:** Note that **xf analysis** cannot be used in this case because we have two outputs (Diff and CM). **xf analysis** is useful for the case of single-ended output op-amp where there is only one output terminal. It is also useful for fully diff op-amp to calculate CMRR (SE2diff), but not Avd/Avcm.



Hand Analysis (A2):

$$A_{vCM} = \frac{v_{oCM}}{v_{iCM}} = \frac{-g_m R_D}{1 + 2(g_m + g_{mb})R_{SS}}, \qquad 2(g_m + g_{mb})R_{SS} \gg 1, \qquad R_{SS} = r_{o,PM3}$$

$$A_{vCM} = \frac{-g_m R_D}{1 + 2(g_m + g_{mb})R_{SS}} = \frac{-(199.689\mu)(25k)}{1 + 2(199.689\mu + 41.7\mu)(\frac{1}{2.4\mu})} = -0.0247 < \mathbf{1}$$

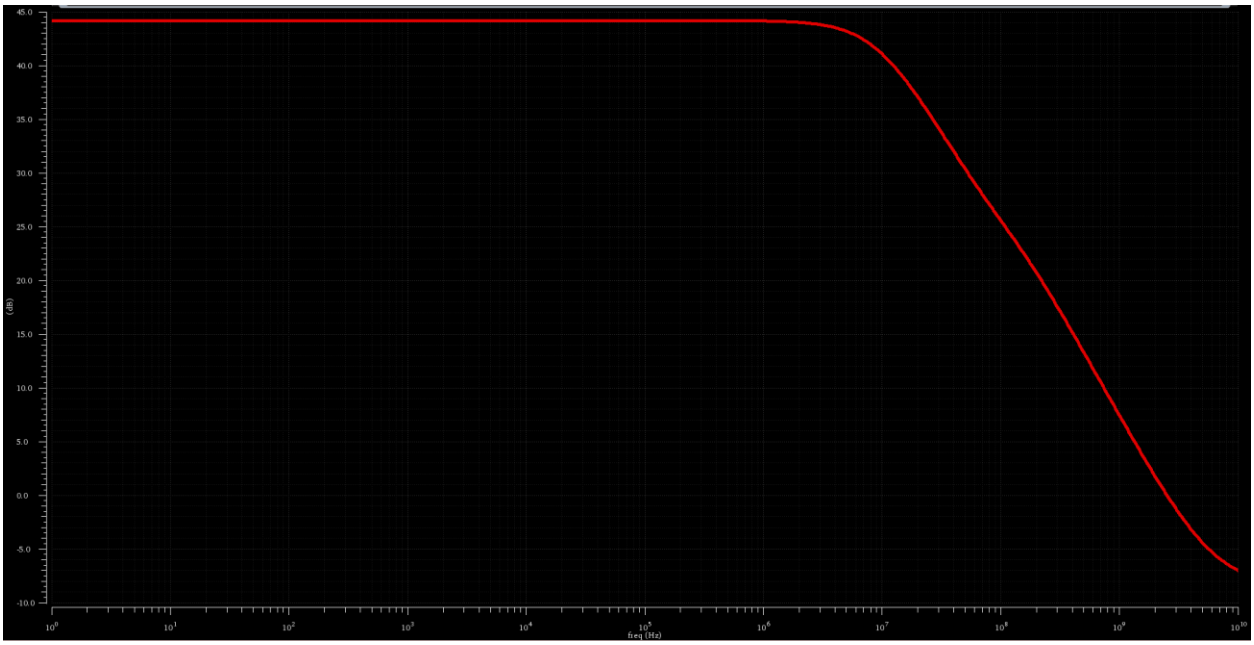
$$20\log|A_{vCM}| = -32.15\ dB$$

	Simulation	Hand Analysis	Relative Error dB (ϵ)
DC CM gain	-30.58 dB	-32.15 dB	-4.76 %

The CM gain $\boldsymbol{A_{vCM}}$ is less than 1 because $\mathbf{2(g_m + g_{mb})R_{SS} \gg 1}$, and CM input is partially rejected.

The variation of **Avcm vs. frequency** shows attenuation of small signal variations in common mode input. At higher frequencies common mode is rejected further more.

Avd/Avcm in dB



$$\frac{A_{vd}}{A_{vCM}} = \frac{-g_m R_D}{\left(\frac{-g_m R_D}{1 + 2(g_m + g_{mb})R_{SS}}\right)} = 1 + 2(g_m + g_{mb})R_{SS} = \frac{-4.9922}{-0.0247} = 202.1133\ V/V$$

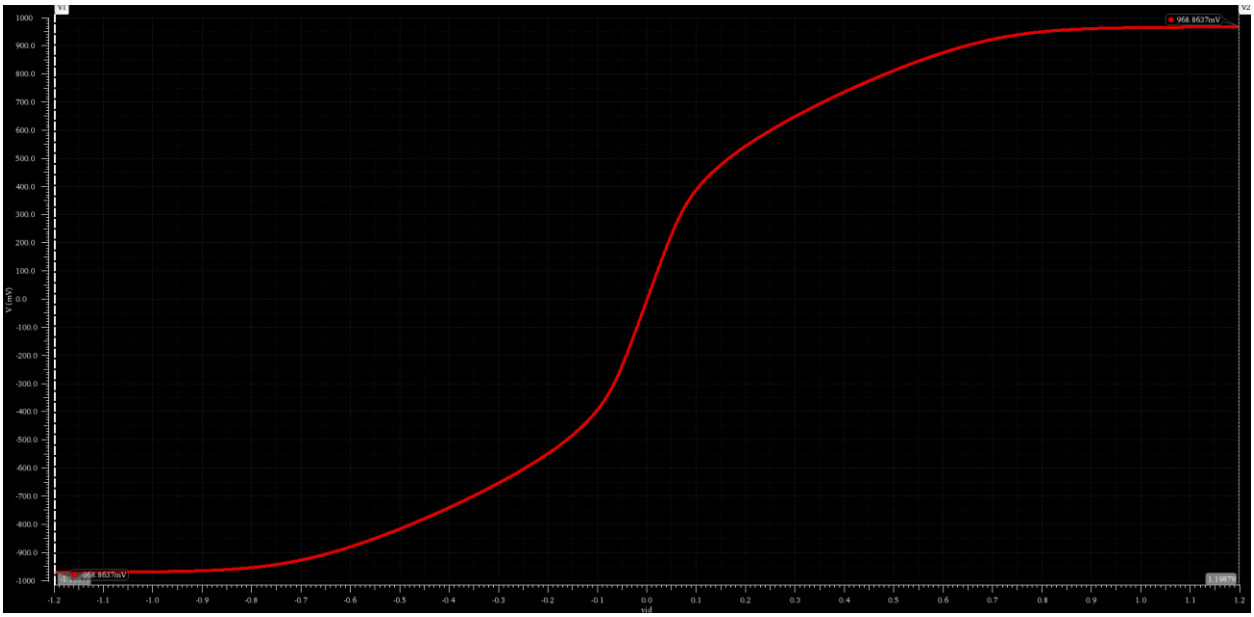
$$20 \log \left|\frac{A_{vd}}{A_{vCM}}\right| = 46.112\ dB$$

	Simulation	Hand Analysis	Relative Error dB (ϵ)
Avd/Avcm	44.245 dB	46.112 dB	-4.048 %

The variation of **Avd/Avcm vs. frequency** shows that the ratio between DC diff gain and DC CM gain is large, and this is one of the benefits of using a differential amplifier, which is Common Mode Rejection.

4) Diff large signal ccs. :

- Use **DC sweep** (not parametric sweep) for **Vid = -VDD: 10m: VDD**. Set Vicm at the center of the CMIR.
- Report diff large signal ccs. (**VODIFF vs. VIDIFF**). Compare the extreme values with **hand analysis** in a table.



Hand Analysis (B1):

$$V_{o,DIFF} = \pm I_{SS}R_D = \pm(40\mu A)(25k\Omega) = \pm 1V$$

	Simulation	Hand Analysis	Relative Error dB (ϵ)
VODIFF vs. VIDIFF extreme values	+/- 968.8637 mV	+/- 1 V	-3.113 %

5) CM large signal ccs. (region vs. VICM):

- We will use the region parameter to know the operating **region of each transistor vs. sweep variable**.

➔ **Cadence Hint:** The "**region**" meaning is as follows: **0: cut-off 1: triode 2: sat 3: subth 4: breakdown**

- Use **DC sweep** (not parametric sweep) for **Vicm = 0: 10m: VDD** (no need to run AC sim.)

➔ **Cadence Hint:** To save the "region" parameter, add this this line to your **save.scs** file (in Model Libraries): **`save *:region sigtype=dev`**.

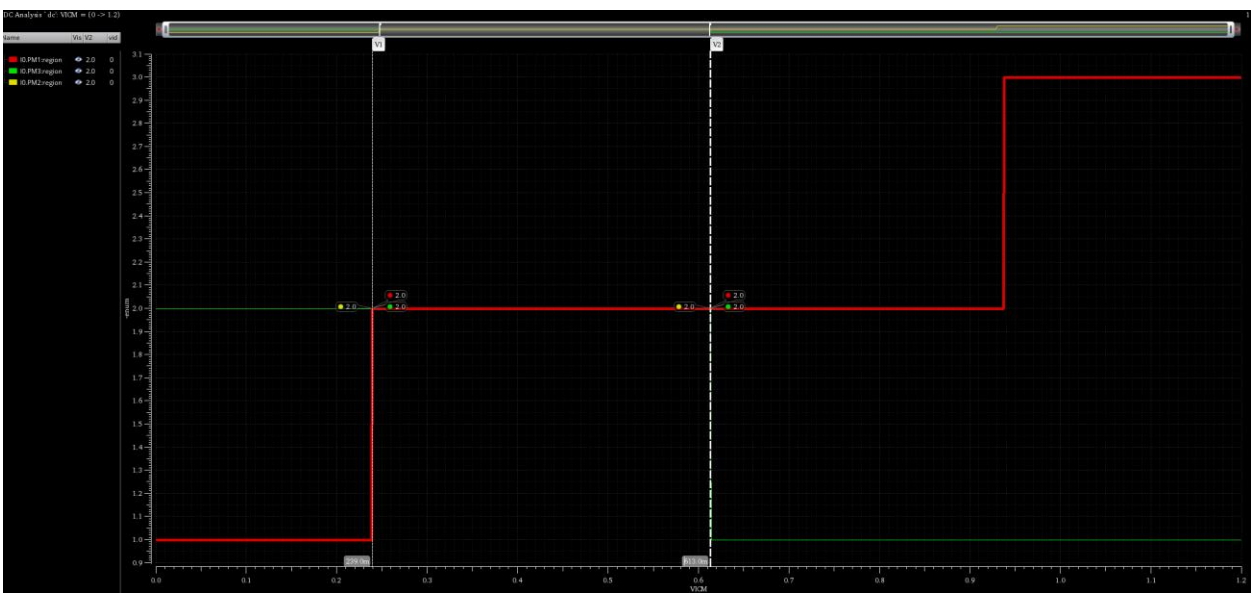
- Disable ac analysis.
- Plot **"region" OP parameter vs. VICM** for the input pair and the tail current source.
- Find the **CM input range (*CMIR*).** Compare with **hand analysis** in a table.

➔ **Note that** the drawback of this method is that the "region" parameter cannot be experimentally measured in the lab and is not quantitatively related to circuit specifications.

PM4 is diode connected, so it is always in saturation.

PM3 is the tail current source. It has a compliance range **from 0 to 613mV**.

PM1 and PM2 are the input pair. They are in saturation **from 239mV to 937mV.**



$$CMIR_{sim} = 613\ mV - 239\ mV = 374\ mV$$

From hand analysis we proved that CMIR is: $243.3\ mV \leq \boldsymbol{V_{iCM}} \leq 571.927\ mV$

	Hand Analysis		Regions Simulation		Relative Error (ϵ)
To / From	571.927 mV	243.3 mV	613 mV	239 mV	+13.81 %
CMIR	328.627 mV		374 mV		

6) CM large signal ccs. (GBW vs. VICM):

- Use **AC analysis** (start = 1, stop = 1, pts(linear) = 1) to get **Avd**. Use **parametric sweep** (not dc sweep) for **Vicm = 0:20m:VDD**.

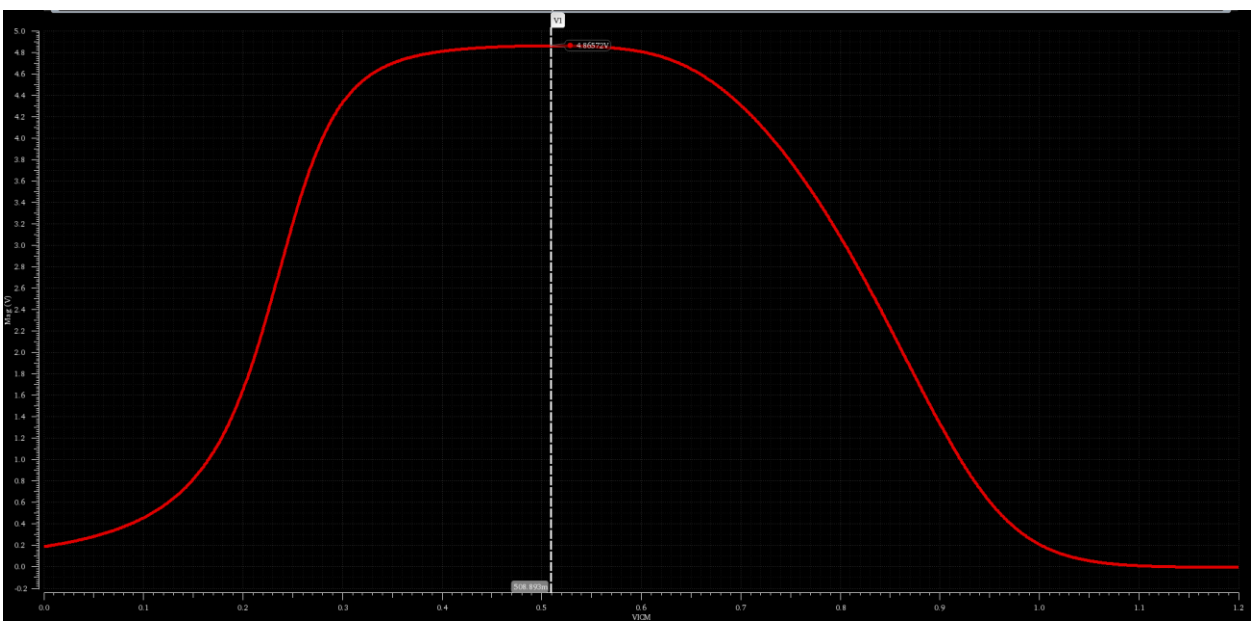
➔ **Cadence Hint:** Instead of using parametric sweep, a better alternative in Cadence is to **use AC sweep but sweep a design variable (Vicm) instead of sweeping the frequency**. The frequency is set at 1 Hz (or any other small value) to get the low frequency gain (DC gain).

➔ **Cadence Hint:** Use the following expression in adexl to calculate Avd: **`ymax(mag(VF("/VODIFF")))`**

- Note that the bandwidth is determined by RD and CL, thus **Avd variation is itself GBW variation**.

- Report **CM large signal ccs (Avd vs. Vicm).** Assume the valid range for Vicm (CMIR) is defined by the condition that *Avd is within 90% of the max gain, i.e., 10% drop in gain.*
- Plot the results overlaid on the results of the previous method (region parameter). Find the CM input range. Compare with the previous method in a table.

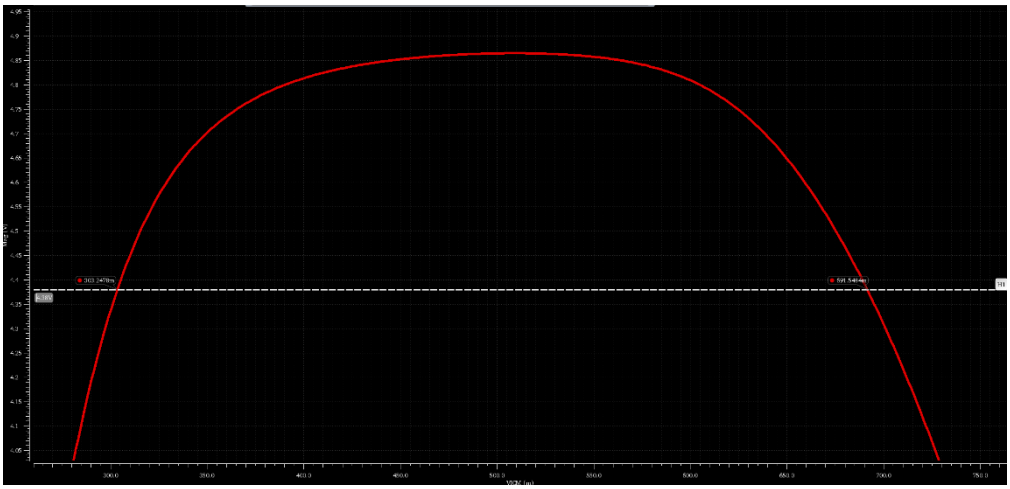
First we need to adjust Vid AC magnitude = 1, and ViCM AC magnitude =0.



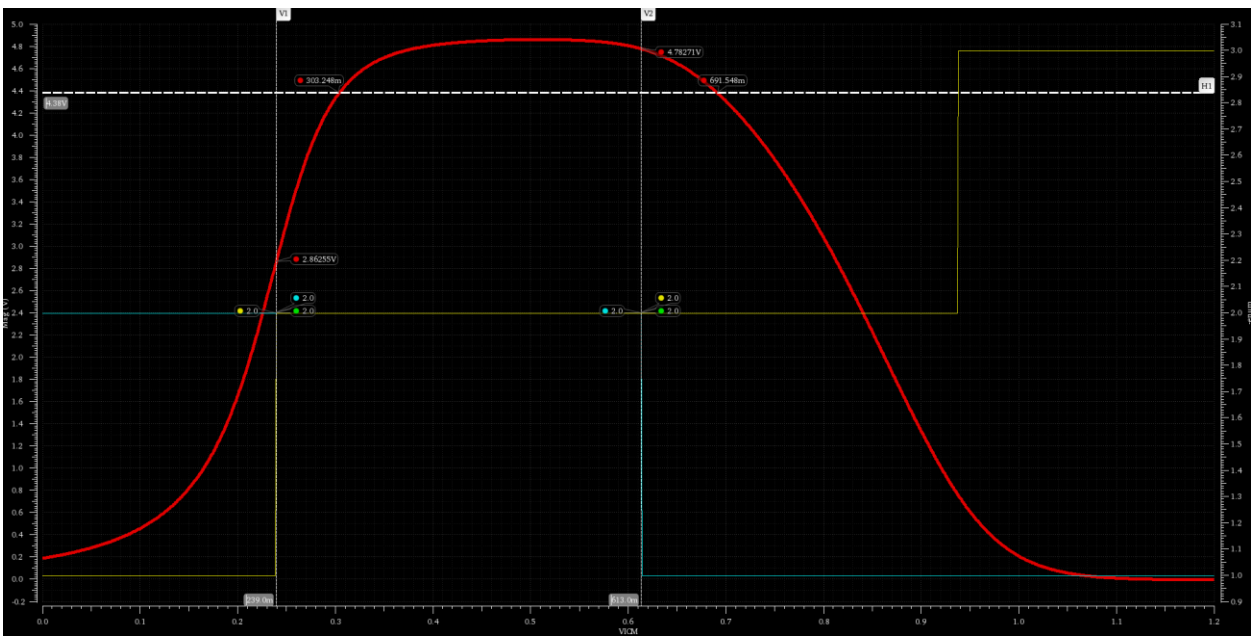
`ymax(mag(VF("/VODIFF")))` = 4.866 V/V

Assuming CMIR = Avd within 90% -> 100% of gain: 4.866*0.9=4.3794 V/V

As shown in the next figure: CMIR = 303.248m -> 691.55m



Plotting the results overlaid on the regions:



	Hand Analysis		Avd (GBW) vs. ViCM Simulation		Regions Simulation	
To / From	571.927mV	243.3 mV	691.548 mV	303.248 mV	613 mV	239 mV
CMIR	328.627 mV		388.3 mV		374 mV	
Relative Error (ϵ) w.r.t Hand Analysis	0 %		+18.15 %		+13.81 %	

Lab Summary

- *In Part 1 I learned*
  - How to generate and use design charts for NMOS and PMOS transistors.
  - How to design a resistive-loaded differential amplifier.

- *In Part 2 I learned*
  - How to use an ideal balun.

- *In Part 3 I learned*
  - How to simulate the small-signal differential gain of a differential amplifier.
  - How to simulate the small-signal common-mode gain of a differential amplifier.
  - How to simulate the large-signal differential characteristics of a differential amplifier.
  - How to simulate the large-signal common-mode characteristics of a differential amplifier.

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[END OF LAB 6]

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